

NANDAN KUMAR

Unity 3D Developer | Extended Reality (XR) / Virtual Reality (VR) / Augmented Reality (AR) | Spatial Computing | C# Scripting

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SUMMARY

Unity 3D Developer specializing in Extended Reality (XR), Virtual Reality (VR), and Augmented Reality (AR) simulation with 10 months of production deployment at Sony SISC. Shipped 4 enterprise XR training modules — industrial, medical, and educational — all sustaining 60 frames per second (FPS) on mid-range hardware with zero missed sprints and zero regression. Achieved 35% draw call reduction, 30% faster scene load times, and 25% memory overhead reduction through GPU Profiling, Object Pooling, and Level of Detail (LOD) optimization. Full stack across OpenXR, XR Interaction Toolkit (XRIT), AR Foundation, Meta XR SDK, Leap Motion SDK, Haptic Feedback, Shader Graph, Universal Render Pipeline (URP), High Definition Render Pipeline (HDRP), and C# Scripting — deploying to Meta Quest 2/3, HoloLens 2, WebGL/WebXR, ARCore, and ARKit.

EXPERIENCE

Unity 3D XR Developer | Sony Semiconductor India Software Centre (Sony SISC) — Bengaluru, India July 2025 – Present

- **Shipped 4 production-grade Extended Reality (XR) / Virtual Reality (VR) training modules** — chemical lab simulation, surgical simulator, physics lab, and bio lab — sustaining 60 FPS on mid-range hardware with a zero-blocker release rate across every 2-week Agile/Scrum sprint in a 5-person team.
- **Reduced scene load times 30%** by decomposing monolithic C# scripts into an event-driven modular architecture, removing 40% of redundant Update() calls and recovering critical CPU frame budget for real-time simulation performance.
- **Achieved 35% draw call reduction** by implementing GPU-instanced Particle System Visual Effects (VFX) on a coroutine-driven state machine in Unity 3D, sustaining stable 60 FPS across 4 simultaneous multi-reaction sequences under peak GPU load.
- **Eliminated 25% memory overhead** by deploying Object Pooling and Level of Detail (LOD) Systems across a multi-module XR educational platform, achieving zero Garbage Collection (GC) allocation spikes and consistent frame pacing at simultaneous peak load.
- **Resolved 15+ critical quality assurance (QA) defects** — Rigidbody physics drift, multi-aspect-ratio UI canvas failures, spatial collision edge cases — with zero regression across all subsequent production releases.
- **Architected OpenXR and XR Interaction Toolkit (XRIT) 6 Degrees of Freedom (6DoF) spatial interaction pipelines** into 2 enterprise medical and industrial training simulations, removing traditional input device dependency for institutional end users across 4 deployment targets.

PROJECTS

XR Industrial Chemical Lab Simulation — Unity 3D | OpenXR | XR Interaction Toolkit (XRIT) | Universal Render Pipeline (URP) | Shader Graph | Particle System | C#

- **Reduced draw calls 35%** by building GPU-instanced Particle System VFX (color change, gas emission, bubbling) on a coroutine-driven reaction state machine in Unity 3D, maintaining stable 60 frames per second (FPS) across simultaneous reaction sequences in industrial Virtual Reality (VR) training deployment.
- **Delivered touchless 6 Degrees of Freedom (6DoF) equipment manipulation** by integrating OpenXR Input System and XR Interaction Toolkit (XRIT) spatial interactions, replacing physical input devices and permitting zero-friction trainer onboarding.

Screw Insertion Surgical Simulation — Unity 3D | Rigidbody Physics | Haptic Feedback | Mixed Reality (MR) | HoloLens 2 | C# Scripting

- **Shipped physics-accurate surgical Virtual Reality (VR) training to medical institutions** using Rigidbody torque mechanics with configurable bone density, insertion-angle constraints, and Haptic Feedback integration — delivering measurable, repeatable procedural training outcomes.
- **Implemented Mixed Reality (MR) overlay via HoloLens 2-compatible pipeline** layering virtual surgical guidance onto physical workspace context, improving trainee spatial awareness and reducing procedural orientation errors.

XR Bio Lab — Persistent Augmented Reality (AR) — Unity 3D | OpenXR | AR Foundation | ARCore | ARKit | Object Pooling | Level of Detail (LOD) | C#

- **Cut memory overhead 25%** via Object Pooling and Level of Detail (LOD) Systems across Meta Quest 2/3, ARCore, and ARKit — smooth cross-device Extended Reality (XR) performance with zero frame-budget compromise; AR Foundation spatial anchors slashed session-setup friction to zero.

Leap Motion Hand Interaction — Digital Twin — Unity 3D | Leap Motion SDK | Gesture Recognition | Hand Tracking | C# Scripting

- **Eradicated 20ms input latency** by optimizing the Leap Motion SDK tracking pipeline for Gesture Recognition-driven grab/release interactions; 1:1 Digital Twin hand mirroring facilitating high-fidelity Virtual Reality (VR) operator training without physical lab access.

Physics Lab Simulation — WebGL / WebXR — *Unity 3D | Raycasting | Line Renderer | WebGL | WebXR Device API | C#*

- **Improved laser simulation accuracy 40%** via dynamic raycasting with multi-surface collision handling; WebGL / WebXR deployment expanded platform access to 100% of browser-accessible institutional devices — zero additional hardware required.

SKILLS

Extended Reality (XR) / Virtual Reality (VR) / Augmented Reality (AR): OpenXR, XR Interaction Toolkit (XRIT), AR Foundation, Meta XR SDK, Oculus SDK, ARCore, ARKit, Leap Motion SDK, Vuforia, 6 Degrees of Freedom (6DoF), Mixed Reality (MR), Haptic Feedback, Gesture Recognition, Hand Tracking, Spatial Computing, Digital Twin

Game Engine & Rendering: Unity 3D (2D/3D), Universal Render Pipeline (URP), High Definition Render Pipeline (HDRP), Shader Graph, VFX Graph, Particle System, Cinemachine, Animation Rigging, Line Renderer, WebGL, WebXR

Hardware Platforms: Meta Quest 2, Meta Quest 3, Oculus Quest 2/3, HTC Vive, HoloLens 2, Android XR, iOS XR, PC VR, WebGL Browser, WebXR

Performance Optimization: GPU Profiling, CPU Profiling, Unity Profiler, Object Pooling, Draw Call Batching, Level of Detail (LOD) Systems, Occlusion Culling, Garbage Collection (GC) Management, 60 FPS Optimization, Frame-Rate-Independent Logic

Software Architecture: C# Scripting, Object-Oriented Programming (OOP), Event-Driven Systems, Coroutines, Scriptable Objects, State Machines, Modular Design, Design Patterns, Agile/Scrum (2-week sprints)

Tools & Version Control: Git, GitHub, Visual Studio, REST API, Spring Boot, MySQL, Java, Postman

Expanding: Photon PUN2 (Multiplayer Networking), Unity Netcode for GameObjects, Vuforia Image Tracking, WebXR Device API, Three.js, Unreal Engine (Blueprints)

EDUCATION

Bachelor of Engineering (B.E.) — Computer Science

August 2020 – June 2024

Chandigarh University, Punjab, India